



FIRST SEMESTER 2026-27
COURSE HANDOUT (Part - II)

Date: 03.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : BITS F218
Course Title : General Mathematics III
Instructor –in –Charge : BHUPENDRA KUMAR SHARMA
bksharma@pilani.bits-pilani.ac.in, Chamber No: 6067G

1. Course Description: Linear equations and matrices, Determinants, Basis of R^n , Eigen Value, Eigen Vector, Linear transformations on R^n . Linear Programming: Geometric Solutions, Simplex Method, Duality, Post optimal Analysis, Transportation and Assignment Problem. Nonlinear Programming (Unconstrained optimization).

2. Scope and Objective of the Course:

The course is made for Pharmacy students keeping in mind the importance of Linear Algebra and Linear programming problems and their possible applications in various fields of science and engineering. Linear algebra begins with vectors and matrices – two of the most important practical concepts in mathematics. Whereas Linear programming is a method to achieve the best outcome in a mathematical model whose requirements are represented by linear relationships. Linear programming can be applied to various fields of study. It is used in business and economics, but can also be utilized for some engineering problems. Industries that use linear programming models include transportation, energy, telecommunications, and manufacturing.

3. Course learning outcomes:

CLO1. Apply the concepts of matrix algebra, systems of linear equations, vector spaces, linear transformations, and eigenvalues/eigenvectors to solve mathematical problems arising in science and engineering.

CLO2. Formulate and solve linear programming problems using graphical and simplex-based methods, analyze duality and post-optimality, and apply optimization techniques to transportation and assignment problems.

CLO3. Analyze and solve basic nonlinear optimization problems using appropriate numerical optimization techniques, and interpret the results in the context of engineering and pharmaceutical applications.

4. Text Books:

T1: E. Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley, 2015.

T2: HA Taha, Operation Research: An Introduction, Pearson Education, 9/E, 2011.

Reference Books:

R1: S. Andrilli and D. Hecker, Elementary Linear Algebra by, 3rd Edition, Elsevier, 2006.

R2: SS Rao, Engineering Optimization: Theory and Practice, New Age International (P) Limited, 3rd Edition, 1996.

R3: BS, Grewal, Higher Engineering Mathematics, 40th Edition, Khanna Publication, 2009.

5. Course Format & Weekly Workflow: weekly schedule for 14 weeks with [3] lectures per week



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Course Plan:

Module No.	Lecture Session/Tutorial Session	Reference	Learning outcomes
1. Matrix Algebra	L1.1-L1.3: Matrix addition, multiplication and transpose T: Discussion of related exercise problems.	T1 (7.1-7.2)	1. Perform standard operations with matrices including addition, scalar multiplication, etc. 2. Write down the coefficient matrix and augmented matrix of a linear system 3. Use elementary row operations to reduce matrices to echelon forms
	L1.4-L1.6: Linear system of equations and row equivalent forms - matrices, augmented matrices, echelon form, Gauss elimination	T1 (7.3)	
	T: Discussion of related exercise problems	T1 (7.4)	4. Understand the meaning and determine whether sets of vectors are linearly independent or dependent
	L1.7-L1.8: Linear dependence, rank of matrix T: Discussion of related exercise problems		
	L1.9-L1.11: Solution of linear system: homogeneous & Non-homogeneous system, determinant, adjoint matrix, Cramer's rule, inverse matrix: Gauss Jordan method T: Discussion of related exercise problems.	T1 (7.5-7.8)	1. Make use of echelon forms in finding the solution sets of linear systems 2. Identify a system of linear equations (or linear system) and describe its solution set
2. Vector space	L2.1-L2.3: Basis and dimensions, linear transformation of R^n . T: Discussion of related exercise problems.	T1 (7.9)	Perform standard operations with vectors in Euclidean space
	L2.4-L2.5: Eigenvalues & eigenvectors T: Discussion of related exercise problems.	T1 (8.1)	Find eigenvalues and eigenvectors of square matrices
3. Linear programming problem	L3.1-L3.5: Introduction to LP – formulation to LPP, graphical solutions T: Discussion of related exercise problems.	T2 (2.1, 2.2, 2.4)	1. Obtain an overview of the kinds of problems linear programming has been used to solve. 2. Learn how to develop linear programming models for simple problems.
	L3.6-L3.10: Algebraic solutions - Simplex method, Artificial variable method, Special cases T: Discussion of Related Exercise Problems.	T2 (3.1-3.5)	1. Be able to identify the special features of a model that make it a linear programming model. 2. Learn how to solve linear programming models by the Simplex method procedure.
	L3.11-L3.14: Duality and dual simplex method T: Discussion of related exercise problems.	T2 (4.1, 7.4, 4.4.1)	Understand how alternative optimal solutions, infeasibility and unboundedness can occur in linear programming problems



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	L3.15: Introduction to post optimal analysis	T2 (4.5)	
4. Transportation and assignment problem	L4.1-L4.5: Transportation and Assignment problem, Finding BFS, Using North-West Corner Rule, LCM & VAM, Modi's method, Hungarian Method for solving assignment problem. T: Discussion of related exercise problems.	T2 (5.1, 5.3, 5.4)	Understand the importance of Transportation and Assignment problem and how to solve such problems.
5. Non-linear programming	L5.1-L5.4: Single variable, Unconditional optimization, Fibonacci search method, Golden Section Search Method, Gradient based method T: Discussion of Related Exercise Problems.	T2 (5.7) T1 (21.1.2)	To understand the problems related to Non-linear programming.

6. Evaluation Scheme:

Component	Weightage (%)	Duration	Date & Time	Remarks
Announced Quizzes	15%	20 minutes	To be announced	Best score of 3 quizzes will be considered out of 4 quizzes
Assignment	5%	-	To be announced	
Midsem	35%	90 minutes		Closed Book
Comprehensive	45%	180 minutes		(CB/OB)

7. Criterion for NC: Minimum 10% of Total marks of the course.

8. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

9. Chamber consultation hour: To be announced in the class.

10. Makeup Policy: Makeup will be given only for genuine cases (accompanied by proof) for Mid-sem or comprehensive exam only.

11. Notices: All notices regarding BITS F218 will be displayed on NALANDA.

Instructor-in-charge
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